Discrete Probability Distributions

Grade 10 · Global Economics · Term 2

Model uncertain outcomes, compare return and risk, and justify an investment choice.

From uncertain returns to an investment choice

Step 1 Describe uncertain returns

Let X be Investment A's annual return, recorded as 2, 4 or 6 when expressed as a percentage. These possible values form its **support**. The **probability mass function (PMF)** gives the probability of each exact outcome. The **cumulative distribution function (CDF)** gives the probability that X is at or below a threshold.

C1–C2 · Read exact and cumulative probabilities

Investment A's annual return distribution

Return x **(%)**	**PMF** $P(X = x)$	**CDF** $P(X \leq x)$
2	0.25	0.25
4	0.50	0.75
6	0.25	1.00

All PMF entries are non-negative and $0.25 + 0.50 + 0.25 = 1$. The CDF never decreases. It is 0 below 2, stays flat between outcomes and is 1 at and above 6.

Point	$P(X = 2) = 0.25$
Interval	$P(2 < X \leq 4) = 0.75 - 0.25 = 0.50$
Cumulative	$P(X \leq 4) = 0.75$
Complement	$P(X > 4) = 1 - 0.75 = 0.25$

Criteria for this step

C1 Characterises discrete random variables in economic and financial contexts by identifying their support, probability mass function and possible outcomes.

C2 Interprets PMFs and CDFs by verifying their defining properties and using them to calculate point, interval, cumulative, and complementary probabilities in financial contexts.

Step 2 Count outcomes in fixed trials

For one loan, a Bernoulli indicator is 1 if it defaults within a year and 0 if it does not. "Success" means the event we count, so here it means a default. Let X count defaults across four loans over that year.

C3–C4 · Check the binomial conditions before calculating

Fixed trials	Two outcomes	Model assumptions
$n = 4$ loans	Default or no default per loan	Independent outcomes, with $p = 0.10$ for every loan

For exactly k defaults, the binomial coefficient counts which k of the n loans default.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad P(X = 1) = 4 \times 0.10 \times 0.90^3 = 0.2916.$$

Criteria for this step

C3 Models binary economic outcomes with Bernoulli random variables, interpreting success and failure in contexts such as default, gain, loss, or claim occurrence.

C4 Applies the binomial distribution to fixed-trial financial situations by verifying its conditions and calculating probabilities using the binomial probability formula $P(X = k) = C(n, k) \times p^k \times (1-p)^{n-k}$.

Step 3 Count trials until success or events in an interval

To model sales calls, assume independent trials with the same probability $p = 0.20$ of a sale on each call. Count all calls up to and including the final sale. For rare insurance claims, count events in one week. A Poisson model is reasonable if the rate is stable and events occur independently. Its parameter λ is the expected count in that week.

C5–C7 · What stops the count?

Three models and their probabilities

Model	What k counts	Worked example
Geometric	Calls through the first sale $k \geq 1$	First sale on call 3 $0.80^2 \times 0.20 = 0.128$
Negative binomial	Calls through the rth sale $k \geq r$	Second sale on call 3 $2 \times 0.80 \times 0.20^2 = 0.064$
Poisson	Claims in one week $k = 0, 1, 2, \ldots$	No claims when $\lambda = 2$ $e^{-2} \approx 0.1353$

For the second sale on call 3, exactly one of the first two calls must be a sale. Either call can be that first sale.

Criteria for this step

C5 Models waiting-time situations with geometric distributions, calculating the probability of the first success occurring on a given trial in financial or business contexts.

C6 Applies the negative binomial distribution to situations that count the number of trials needed to obtain a fixed number of successes.

C7 Evaluates rare-event situations with the Poisson distribution, interpreting the parameter λ in contexts such as claims, arrivals, defects, or defaults within a fixed interval.

Step 4 Measure return and risk, then choose a model

Expected return is the long-run average over repeated years under the same distribution. **Variance** measures spread around that average in squared units. To choose a probability model for a new situation, ask what the variable counts and check the assumptions.

C8–C9 · Use Investment A's PMF, then match a model to a new count

$$\begin{aligned} &\textbf{Mean} && 0.25 \times 2 + 0.50 \times 4 + 0.25 \times 6 = 4 \\ &\textbf{Variance} && 0.25(2-4)^2 + 0.50(4-4)^2 + 0.25(6-4)^2 = 2 \end{aligned}$$

Weight each outcome by its probability and add to get an expected return of 4%. Weight the squared distances from 4 in the same way to get a variance of 2 squared percentage points.

Model selection guide

Model	What it counts	What to check
Bernoulli	Success (1) or failure (0)	One trial, two outcomes, success probability p
Binomial	Successes in n trials	Fixed n, independent binary trials, constant p
Geometric	Trials through the first success	Independent binary trials, constant $p > 0$
Negative binomial	Trials through the rth success	Fixed target r, independent binary trials, constant $p > 0$
Poisson	Events in a fixed interval	Stable rate, independent event occurrences

Choose a model. A lender counts defaults among 20 loans over a year. With two outcomes per loan, independent defaults and the same default probability for every loan, use a binomial model for the number of defaults in the fixed group.

Criteria for this step

C8 Calculates and interprets the expected value and variance of discrete probability distributions as measures of return and risk in economic and financial contexts.

C9 Compares Bernoulli, binomial, negative binomial, geometric, and Poisson distributions by selecting the most appropriate model for a given financial-risk situation.

Step 5 Justify an investment choice

Screen two hypothetical investments for inclusion in a portfolio over the same one-year horizon. Compare each investment's expected return with its spread and the possibility of a loss.

C10 · A higher expected return comes with a wider spread

Annual returns and risk for two investments

Probability or measure	Investment A	Investment B
0.25	2%	−4%
0.50	4%	6%
0.25	6%	16%
Expected return	4%	6%
Variance (squared percentage points)	2	50

A cautious investor could include A for its smaller variance and positive return in every modelled outcome. An investor willing to accept more risk could include B for its higher expected return, accepting its larger variance and 25% chance of a loss.

Criteria for this step

C10 Assesses investment alternatives for inclusion in a portfolio by comparing their discrete return distributions, expected returns, variances, and risk–return trade-offs, using the results to support investment portfolio decision-making.